


DATA SET
D51405

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.² Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married	
	Yes	No
Bachelors or higher	187	113
Some college	162	138
High school or less	149	151

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

H_0 : marriage rates and education level are independent
 H_a : they are dependent

$$\frac{300 \times 187}{900} = 65.6$$

$$\frac{300 \times 162}{900} = 54$$

$$\frac{300 \times 149}{900} = 50$$

Row totals or higher:

$$\chi^2_{Yes} = \frac{(187-65.6)^2}{65.6} \approx 265.6$$

$$\chi^2_{No} = \frac{(113-134.4)^2}{134.4} \approx 3.241$$

College:

$$\chi^2_{Yes} = \frac{(162-65.6)^2}{65.6} \approx 0.066$$

$$\chi^2_{No} = \frac{(138-134.4)^2}{134.4} \approx 0.009$$

High school or less:

$$\chi^2_{Yes} = \frac{(149-65.6)^2}{65.6} \approx 1.741$$

$$\chi^2_{No} = \frac{(151-134.4)^2}{134.4} \approx 2.157$$

Total $\chi^2 = 60.04$
 $\alpha = 0.01$ df = 2
 $\chi^2_{crit} = 5.991$
 $60.04 > 5.991$, reject H_0 .

DATA SET

D51407

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence		
	Apartment	Duplex	Single Residence
1	8	20	1
2	16	8	9
3	10	10	14
4 or More	6	2	16

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

H_0 : 3 types of residence & family members are independent
 H_a : there is a significant diff

$$\chi^2_{1 \text{ member}} = \frac{24^2}{120} = 4.8$$

$$\chi^2_{2 \text{ member}} = \frac{32^2}{120} = 8.533$$

$$\chi^2_{3 \text{ member}} = \frac{34^2}{120} = 9.633$$

$$\chi^2_{4 \text{ or more}} = \frac{30^2}{120} = 7.5$$

1 member: $\chi^2 = \frac{(8-4)^2}{4.8} \approx 3.33$
 2 members: $\chi^2 = \frac{(16-8.533)^2}{8.533} \approx 6.25$
 3 members: $\chi^2 = \frac{(10-9.633)^2}{9.633} \approx 0.013$
 4 or more: $\chi^2 = \frac{(6-7.5)^2}{7.5} \approx 0.24$

Total $\chi^2 = 9.6$
 $\alpha = 0.01$ df = 3
 $\chi^2_{crit} = 11.345$
 $9.6 < 11.345$, fail to reject H_0 .

19. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1–10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

Meal	A	B	Meal	A	B
1	6	8	12	8	5
2	4	5	13	4	2
3	7	4	14	3	3
4	8	7	15	6	8
5	2	3	16	9	10
6	7	4	17	9	8
7	9	9	18	4	6
8	7	6	19	4	3
9	2	5	20	5	4
10	4	3	21	3	2
11	6	9	22	5	3

- Use the binomial tables in Appendix I to find the exact rejection region for the test.
- Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \geq 25$, it works fairly well for values of n as small as 15.)
- Compare the results of parts a and b.

$n = 22$
 $p = 0.5$
 $H_0: \mu = 0.5$
 $H_a: \mu \neq 0.5$
 $\alpha = 0.05$
 $P(X \leq 11) = 0.0377$
 $P(X \geq 11) = 0.0377$
 $P(X \leq 11 \text{ or } X \geq 11) = 0.0754$
 $\alpha = 0.0754 > 0.05$
 $\therefore$ Fail to reject H_0
 $\therefore$ There is not enough evidence to conclude that one chef tends to give higher ratings than the other.

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs. A. Test using the sign test. Choose your rejection region so that α is near .05.

Hospital	Drug A		
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	84	63	75.0
2	63	44	69.8
3	56	48	85.7
4	77	57	74.0
5	29	20	69.0
6	48	40	83.3
7	61	42	68.9
8	45	35	77.8
9	79	57	72.2
10	62	48	77.4

Hospital	Drug B		
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

$p = 0.5$
 $H_0: \mu = 0.5$
 $H_a: \mu \neq 0.5$
 $\alpha = 0.05$
 $P(X \leq 11) = 0.0377$
 $P(X \geq 11) = 0.0377$
 $P(X \leq 11 \text{ or } X \geq 11) = 0.0754$
 $\alpha = 0.0754 > 0.05$
 $\therefore$ Fail to reject H_0
 $\therefore$ There is not enough evidence to conclude that one drug tends to give higher recovery rates than the other.